



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

MIDTERM EXAMINATION — Fall 2022

COURSE NAME:

Circuit & Lab

COURSE NUMBER:

CSE2120

EXAMINATION DATE:

XX-10-2022

TIME:

EXAMINATION DURATION:

60 minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

Calculators are NOT allowed.

(Do not use)	(Do not use)	(Do not use)	(Do not use)	(Do not use)

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write ID number in each page of your examination paper.

All answers should be written in the space provided for each question, unless specified the other way.

If you have a problem please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

ID () Name ()

Answer Sheet

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

[1-5] Fill in the blanks with appropriate words.

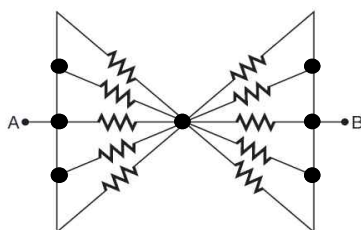
1. In an electric circuit the algebraic sum of () at a node is zero.
2. In an electric circuit the algebraic sum of () in a loop is zero.
3. Node analysis is used to solve () variables of a circuit.
4. The () of an inductor is continuous.
5. While finding response due to one source, all other ideal () sources are replaced with a short circuit.

[6-10] State whether the following statements are True [T] or False [F].

6. In parallel connected elements the voltage will be the same.
7. In an electric circuit, a current flows through short-circuited terminals.
8. An ideal voltage source can be converted into an ideal current source.
9. For node analysis, any node can be chosen as the reference node.
10. In a capacitance, the voltage at $t = 0^-$ is not equal to the voltage at $t = 0^+$.

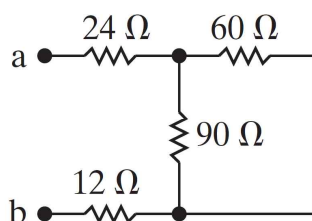
[11-13] Calculate equivalent resistances.

11. In the network shown, if the value of all the resistances are $1\ \Omega$ then what will be the equivalent resistance at A-B?



- (a) $0.2\ \Omega$
- (b) $0.4\ \Omega$
- (c) $0.5\ \Omega$
- (d) $1.0\ \Omega$

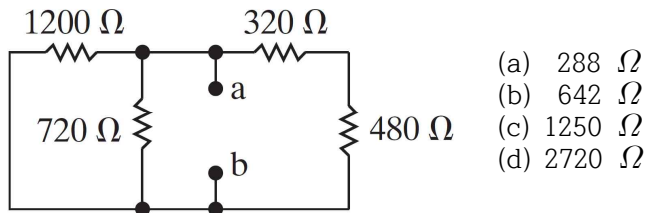
12. Find the equivalent resistance at a-b.



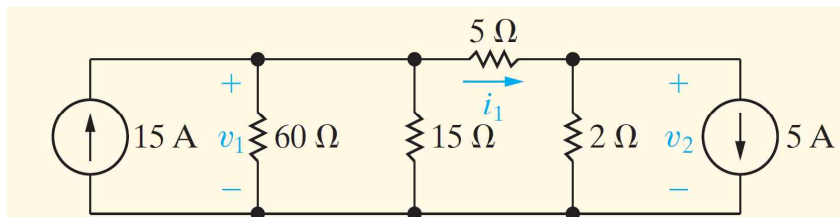
- (a) $36\ \Omega$
- (b) $70\ \Omega$
- (c) $72\ \Omega$
- (d) $186\ \Omega$

ID () Name ()

13. Find the equivalent resistance at a-b.

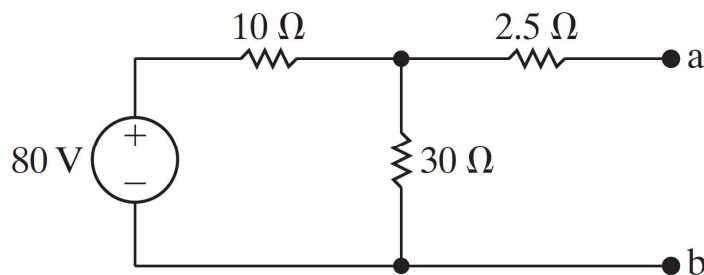


[14-16] In the circuit,



14. Find v_1
 (a) 4 V (b) 60 V (c) 180 V (d) 900 V
15. Find v_2
 (a) 10 V (b) 25 V (c) 75 V (d) 180 V
16. Find i_1
 (a) 1 A (b) 3 A (c) 5 A (d) 10 A

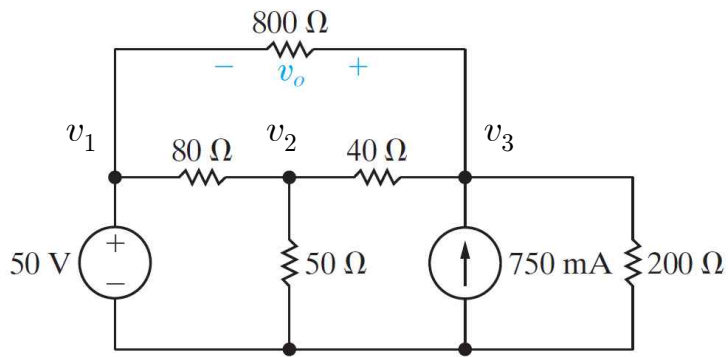
[17-20] Find the Thevenin equivalent with respect to the terminals a-b.



17. What is the Thevenin voltage, V_{TH} ?
 (a) 10 V (b) 30 V (c) 60 V (d) 80 V
18. What is the Thevenin resistance, R_{TH} ?
 (a) 7.5 Ω (b) 10 Ω (c) 12.5 Ω (d) 32.5 Ω
19. When 10 Ω resistor is connected between a and b, what is the voltage across this 10 Ω ?
 (a) 5 V (b) 7.5 V (c) 15 V (d) 30 V
20. When 5 Ω resistor is connected between a and b, what is the voltage across this 5 Ω ?
 (a) 2.5 V (b) 7.5 V (c) 20 V (d) 30 V

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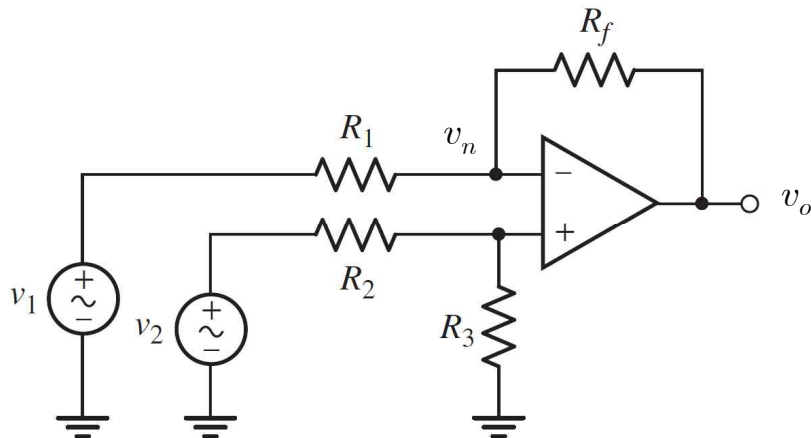
[21-24] When the bottom nodes are the reference nodes in the circuit,



21. Find the node voltage v_1 .
(a) 23.5 V (b) 30 V (c) 37.4 V (d) 50 V
22. Find the node voltage v_2 .
(a) 34 V (b) 48 V (c) 50 V (d) 55 V
23. Find the node voltage v_3 .
(a) 12.2 V (b) 30 V (c) 32.5 V (d) 53.2 V
24. Find v_o .
(a) 2.5 V (b) 3.2 V (c) 5 V (d) 6.5 V

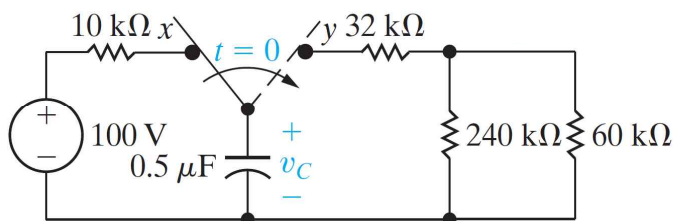
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[25-26] When $R_1 = R_2 = R_3 = 1\text{ k}\Omega$, $R_f = 2\text{ k}\Omega$, $v_1 = 6\text{ V}$, and $v_2 = 4\text{ V}$;



25. Find v_n .
 (a) 0 V (b) 2 V (c) 4 V (d) 6 V
26. Find v_o .
 (a) -6 V (b) -4 V (c) -2 V (d) 0 V

[27-30] Determine the following:



27. Find $v_C(0^-)$
 (a) 0 V (b) 10 V (c) 100 V (d) 200 V
28. Find $v_C(\infty)$
 (a) 0 V (b) 10 V (c) 100 V (d) 200 V
29. Find the time constant for $t \geq 0$.
 (a) 2.5 ms (b) 5 ms (c) 16 ms (d) 40 ms
30. Find $v_C(0^+)$
 (a) 0 V (b) 10 V (c) 100 V (d) 200 V